

Specific-Referenced Indefinite Quantifiers as Ordinal Refiners: A RED-Based Operator Extension of Cardinal–Ordinal Correlation

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Abstract

Cardinals (three) measure quantity and ordinals (third) identify positions in an ordered domain, yet natural language permits systematic shifts between these regimes. Building on a bidirectional discretization architecture in which an ordinalization operator R refines coarse quantity into addressable order positions, and inverse unification operators collapse refined microstructure back into macro-level wholes, this article extends the program to a third layer: specific-referenced indefinite quantifiers (e.g., all, both, either, neither, certain, half, exact, the least, the most). The central claim is that these quantifiers function as higher-order ordinal refiners: given an ordinal spine O , they select recoverable blocks or alternative sets over O , preserving enough ordinal identity to support anaphora, ellipsis, contrast, and controlled reverse entailment. Crucially, because the RED methodology is bidirectional, ordinal refinement is not the end-state: licensed collapse maps allow a quantifier-refined ordinal structure to feed back into cardinal inferences (lower bounds, bounded choice summaries, extremal summaries) in a principled way. We formalize this interaction as a composed pathway, $\text{Cardinal} \rightarrow \text{Ordinal} \rightarrow \text{SR-Quantifier refinement} \rightarrow \text{Collapse} \rightarrow \text{Cardinal summary}$, and we introduce RED-QORD (Reverse Entailment Diagnostic for quantifier–ordinal interaction) to classify uses as REFINING vs COARSENING depending on recoverability of ordinal traceability under paraphrase and ellipsis. Two worked case studies (English and German) illustrate the end-to-end protocol and motivate a compact set of distinguishing predictions against lexical-derivational and rank-only baselines. We conclude with implications for neurolinguistic processing (ordinal traceability as a memory-retrieval constraint) and for computational semantics and information extraction (recovering ordered subset structure from quantificational expressions). (Ionin & Matushansky 2006; Nouwen 2009; Alstott 2024; Barwise & Cooper 1981; Keenan & Stavi 1986).

Box 1. Novelty and Contributions

This article advances a testable operator-diagnostic account of how specific-referenced indefinite quantifiers (SR-indefinites) act as ordinal refiners and feed back into cardinal summaries under RED/DED.

Primary contributions (standard reviewer taxonomy):

1. New diagnostic: RED-QORD classifies quantifier–ordinal constructions as REFINING vs COARSENING via bidirectional entailment and recoverability diagnostics (ellipsis/anaphora, paraphrase, reverse entailment).
2. New composed mapping: a structured pathway Cardinal → Ordinalization → SR-refinement (QORD) → DED-collapse → Cardinal summary, with explicit baselines and failure modes.
3. New measurable parameter: Ordinal Traceability Advantage (OTA), operationalized as a retrieval-continuation advantage in the Minimal Pair Diagnostic Set (MP1–MP4).
4. Cross-domain link: ordinals ↔ SR-quantifiers ↔ cardinal feedback mapping, separating the proposal from (i) lexical-derivational ordinal accounts and (ii) discourse-representation approaches per se.

Related Work and Positioning

The proposal triangulates three bodies of work. First, it is compatible with formal semantics of numerals and type-shifting, where cardinal expressions compose with NP denotations and can enter nominal environments under standard type-shifts. Second, it connects with analyses of ordinals as rank-related expressions (including their interaction with superlative structure), and with morphological/lexical accounts that treat ordinals as derived adjectival material. Third, it aligns with generalized-quantifier semantics in treating determiners/quantifiers as operators over domains, but it makes an additional parameter explicit and testable: whether a quantifier use preserves ordinal identity (recoverable positions/alternatives/blocks) or erases it (yielding only graded magnitude). The distinguishing move is to relocate explanation from morphology alone to an operator-level refinement/collapse interface: (i) ordinals are not merely derived adjectives but can be packaged as addressable NP-like objects, and (ii) specific-referenced indefinite quantifiers can act as refiners over those ordered objects. For German, the account leverages the independently motivated fact that reified ordinals (der/die Dritte) pattern with nominalized adjectives, providing overt diagnostics for NP-like packaging. (Barwise & Cooper 1981; Keenan & Stavi 1986; Partee 1987; Ionin & Matushansky 2006; Nouwen 2009; Alstott 2024; Olsen 1988; Bücking 2025).

1. Introduction: from cardinal–ordinal correlation to quantifier–ordinal–cardinal feedback

Cardinal–ordinal correlation is often described as a close lexical relation inside the numeral system: ordinals are morphologically derived from cardinals and interpreted as rank modifiers. However, the empirical distribution is broader. Ordinals can function as DP-internal modifiers (the third book), as predicates (came third), and as NP-like arguments (the third arrived). These patterns indicate that grammars make available multiple packaging regimes, not a single lexical category. The RED-based operator view reframes the cardinal–ordinal link as bidirectional discretization: a forward refinement step builds an ordered spine from a coarse counting basis, while a reverse collapse step allows the resulting ordinal microstructure to be suppressed back to macro-level summaries. This article adds a third, empirically salient step. Once a domain is ordinalized (for example, a sequence of ranked items or an initial segment of runners), specific-referenced indefinite quantifiers can refine the ordinal spine into discourse-relevant suborders: either of the first two options (explicit alternatives), both of the first two options (exhaustive block), the least of the first ten attempts (extremal block), and related expressions. The core analytic question is then not only how these quantifiers refine ordinals, but how such refinement affects cardinal information under the RED/DED procedures. The answer proposed here is compositional: refinement creates recoverable structure; collapse maps determine which parts of that structure survive as cardinal summaries. The result is a unified pathway in which quantifier–ordinal interaction predictably modulates the strength and form of cardinal inferences. (Ionin & Matushansky 2006; Nouwen 2009; Alstott 2024).

2. Indefinite quantifiers: proportional force vs referential specificity

We adopt a two-dimensional typology that classifies indefinite quantifiers by (i) proportional force and (ii) referential specificity. Proportional force partitions quantifiers into a lower group (below half) and an upper group (above half); referential specificity distinguishes quantifiers that encode a discourse-identifiable scope or choice set from those that do not. The present paper focuses on specific-referenced indefinite quantifiers. These expressions can be “indefinite” insofar as they do not always fix an exact cardinality, but they are specific-referenced in encoding an identifiable domain, scope, or choice set that can be indexed by ordered positions.

This property explains their systematic affinity with ordinal expressions and their ability to participate in RED-QORD diagnostics. (Barwise & Cooper 1981; Keenan & Stavi 1986).

Table 1. Two-dimensional typology of indefinite quantifiers (schematic).

Class	Representative items	Natural anchoring bias
Proportional (lower/upper)	less than half, few, fewer/less; majority, most, all (as proportion)	Cardinal basis (set size)
Specific-referenced indefinite	all, both, either, neither, exact, certain, the least, the most	Ordinal basis (choice set / scope)
Non-specific indefinite	some, many, more, few, several, much, little, plenty	Cardinal basis (numerical grounding)

The specific-referenced class is the target of the present extension: it supports ordinal anchoring and, under RED-QORD, predicts recoverable position structure in contexts that license ellipsis, anaphora, and contrast.

3. Operator architecture: refinement, collapse, and RED-QORD

3.1 Ordinalization and collapse. Let I be a base well-order (an index ordinal) and let $\{F_i : i \in I\}$ be a family of nonempty well-ordered fibers. The ordinalization operator R builds a refined order as a lexicographic sum $R(I, \{F_i\}) = \sum_{i \in I} F_i$, creating addressable positions (i, f) . A canonical collapse map c projects $\sum_i F_i$ back to I , $c(i, f) = i$, forgetting internal microstructure. More semantically substantive coarsening operations (folding, group formation, generic unification) package microstructure into macro-level objects while preserving ordinal anchoring where relevant. (Heim & Kratzer 1998).

3.2 Quantifier refinement over ordinal spines. Given an ordinalized domain O with a salient anchored segment (for example, positions $1..n$), a specific-referenced quantifier Q introduces an additional refinement operator Q^R that selects a recoverable suborder or a block family within

O. Choice quantifiers (either/neither) produce an explicit alternative set; duality/totality quantifiers (both/all) produce an exhaustive block; extremal quantifiers (the least/the most) produce a block characterized by an extremal criterion relative to O. In parallel, Q supports licensed collapse maps Q^c that remove fine-grained ordinal identity when the discourse only requires a macro summary (choice, duality, totality, extremality) rather than exact position tracking.

3.3 Cardinal feedback as composed maps. The central new point for publishability is that quantifier refinement affects the cardinal system through the same bidirectional machinery. Define a cardinalization map card that extracts coarse cardinal profiles (lower bounds, bounded-choice summaries, or graded magnitudes) from an ordered representation. Then the full pathway can be represented schematically as:

$$D (\text{domain}) \rightarrow n (\text{cardinal profile}) \rightarrow R(n)=O (\text{ordinal spine}) \rightarrow Q^R(O) (\text{quantifier-refined blocks/alternatives}) \rightarrow Q^c(Q^R(O)) (\text{macro summary}) \rightarrow \text{card}(\cdot) (\text{cardinal inference}).$$

Under this view, the difference between 'either of the first two options' and 'many options' is not merely lexical: the former creates a recoverable alternative structure anchored to an ordinal segment, which supports a stable lower-bound cardinal inference ($|O| \geq 2$) and licensing conditions for ellipsis/anaphora; the latter typically lacks an ordinal anchor and therefore collapses directly to graded magnitude without recoverable ordinal identity.

3.4 RED-QORD (Reverse Entailment Diagnostic for quantifier–ordinal interaction). RED-QORD adapts RED to the quantifier-ordinal interface by inserting a quantifier stage between ordinalization and collapse. Given a base domain D and a cardinal profile, RED-QORD proceeds as follows: (i) construct an ordinal spine via R ; (ii) apply Q^R to derive a quantifier-indexed suborder or block family; (iii) test reverse entailment and recoverability by attempting controlled collapse (via c and/or Q^c) and checking whether the resulting paraphrase preserves, partially preserves, or eliminates ordinal traceability. The diagnostic returns REFINING when positional identity remains recoverable (for example, explicit alternatives support later retrieval under ellipsis: 'Either of the first two options is acceptable; the first is cheaper, the second is faster'). It returns COARSENING when only graded magnitude or a non-addressable summary remains.

4. Constructions and a fixed elicitation grid

To make the proposal testable across languages and constructions, we adopt a fixed elicitation grid. The grid controls three factors: (i) ordinal anchoring present vs absent, (ii) quantifier class (specific-referenced vs non-specific), and (iii) construction type (DP-internal modifier, reified NP, predicative, iteration). The goal is to separate the contribution of ordinalization from general nominalization or free adjunction effects. In addition to the numeral grid (cardinal counting, ordinal modifier, ordinal nominalization, copular predicate, iteration), we add three quantifier-ordinal cells: choice (either/neither), duality/totality (both/all), and extremal (least/most), each in an anchored 'first n' configuration. The corresponding RED prompt tests the strength of de-ordinalization: if there is an nth (or if the first n are evaluated), then the domain cardinality is at least n.

Table 2. Compact elicitation grid for QORD (illustrative forms; adapt lexemes per language).

QORD cell / construction	English	German	Turkish	Spanish	Russian	Mandarin (Simplified)
Q1. Choice refinement (MP1–MP3) Expected: REFINING DED probe: collapse to {0/1/2}	SR: either/both/neither of the first two N Ctrl: some/many/few of the first two N Retr: The first N..., the second N... DED: So: (0/1/2) of the first two?	SR: entweder die erste oder die zweite der ersten zwei N / beide der ersten zwei N / weder die erste noch die zweite Ctrl: einige/viele/wenige der ersten zwei N Retr: Die erste..., die zweite... DED: Also: (0/1/2) der ersten zwei?	SR: ilk iki N'den ya biri ya diğeri / ilk ikisinin ikisi de / ilk ikisinden hiçbir Ctrl: ilk iki N'den bazıları/çoğu/azı Retr: Birincisi..., ikincisi... DED: Yani ilk ikiden (0/1/2)?	SR: uno de los dos primeros N / ambos de los dos primeros N / ninguno de los dos primeros N Ctrl: algunos/muchos/pocos de los dos primeros N Retr: El primero..., el segundo... DED: O sea: (0/1/2) de los dos primeros?	SR: либо один из первых двух N / оба из первых двух N / ни один из первых двух N Ctrl: некоторые/многие/немногие из первых двух N Retr: Первый..., второй... DED: Значит: (0/1/2) из первых двух?	SR: 前两个N中的任意一个 / 前两个N中的两个都 / 前两个N中一个也不 Ctrl: 前两个N中的一些很多/很少 Retr: 第一个..., 第二个... DED: 也就是说: 前两个里 (0/1/2)?
Q2. Totality on bounded	SR: all of the first three N	SR: alle der ersten drei N	SR: ilk üç N'nin hepsi	SR: todos los tres primeros N	SR: все из первых трёх N	SR: 前三个N全部 Ctrl:

ordinal domain (MP4) Expected: REFINING DED probe: collapse to 3	Ctrl: many of the first three N Retr: The first..., the second..., the third... DED: So: three of the first three?	Ctrl: viele der ersten drei N Retr: Die erste..., die zweite..., die dritte... DED: Also: drei von den ersten drei?	Ctrl: ilk üç N'nin çoğu Retr: Birincisi..., ikincisi..., üçüncüsü... DED: Yani ilk üçten üçü mü?	Ctrl: muchos de los tres primeros N Retr: El primero..., el segundo..., el tercero... DED: O sea: tres de los tres primeros?	Ctrl: многие из первых трёх N Retr: Первый..., второй..., третий... DED: Значит: три из первых трёх?	前三个N中的很多 Retr: 第一个..., 第二个..., 第三个... DED: 也就是说: 前三个=3个?
Q3. Exactness / count refinement (MP5) Expected: REFINING DED probe: collapse to 2	SR: exactly two of the first three N Ctrl: some of the first three N DED: So: two of the first three?	SR: genau zwei der ersten drei N Ctrl: einige der ersten drei N DED: Also: zwei der ersten drei?	SR: ilk üç N'den tam olarak ikisi Ctrl: ilk üç N'den bazıları DED: Yani: ilk üçten iki?	SR: exactamente dos de los tres primeros N Ctrl: algunos de los tres primeros N DED: O sea: dos de los tres primeros?	SR: ровно два из первых трёх N Ctrl: некоторые из первых трёх N DED: Значит: два из первых трёх?	SR: 前三个N中恰好两个 Ctrl: 前三个N中的一些 DED: 也就是说: 前三个里=2个?
Q4. Minimality / extremal low (MP6) Expected: MIXED (context-sensitive) DED probe: elicit k-min	SR: the least of the first ten N Ctrl: few of the first ten N DED: How many (k) are included?	SR: die wenigsten der ersten zehn N Ctrl: wenige der ersten zehn N DED: Wie viele (k) sind es?	SR: ilk on N'nin en azı Ctrl: ilk on N'nin azı DED: Kaç tane (k)?	SR: la menor cantidad de los diez primeros N Ctrl: pocos de los diez primeros N DED: ¿Cuántos (k)?	SR: наименьшее число из первых десяти N Ctrl: мало из первых десяти N DED: Сколько (k)?	SR: 前十个N中最少的 Ctrl: 前十个N中很少 DED: 具体是多少 (k)?
Q5. Maximality / extremal high (MP7) Expected: MIXED (context-sensitive) DED probe: elicit k-max	SR: the most of the first ten N Ctrl: many of the first ten N DED: How many (k) are included?	SR: die meisten der ersten zehn N Ctrl: viele der ersten zehn N DED: Wie viele (k) sind es?	SR: ilk on N'nin en çoğu Ctrl: ilk on N'nin çoğu DED: Kaç tane (k)?	SR: la mayor cantidad de los diez primeros N Ctrl: muchos de los diez primeros N DED: ¿Cuántos (k)?	SR: наибольшее число из первых десяти N Ctrl: многие из первых десяти N DED: Сколько (k)?	SR: 前十个N中最多的 Ctrl: 前十个N中很多 DED: 具体是多少 (k)?
Q6. Non-specific controls	Ctrl: some/many/few of the first	Ctrl: einige/viele/wenige der	Ctrl: ilk N'den bazıları/çoğu/azı	Ctrl: algunos/muchos/pocos de los	Ctrl: некоторые/многие/не многие из первых N	Ctrl: 前N个中的一些很多/很少

(baseline) Expected: COARSEN ING / no stable retrieval advantage DED probe: cardinal summary only	N (bounded) No forced retrieval continuation DED: Summarize as a cardinal count	ersten N Keine erzwungene Rang- Retrieval- Folge DED: Kardinale Zusammenfass ung	Zorunlu sıralama yok DED: Kardinal özet	primeros N Sin recuperación ordinal forzada DED: Resumen cardinal	Без обязательного рангового продолжения DED: Кардинальное резюме	不要求序位回收 DED: 只给出数量总结
Q7. Ordinal NP anchoring (traceability frame) Expected: supports retrieval and RED- QORD reverse tests DED probe: map to cardinal and back	Ordinal NP: the first/second/th ird N Retr: The first..., the second... DED: From ordinal list → cardinal summary → recover ordinal identities	Ordinal NP: der/die Erste/Zweite/ Dritte Retr: Die erste..., die zweite... DED: Ordinal → Kardinal → Ordinal (Rückgewinnu ng)	Ordinal NP: birinci/ikinci/üç üncü Retr: Birincisi..., ikincisi... DED: Sıra → sayı → sıra (geri kazanım)	Ordinal NP: el/la primero/a, segundo/a, tercero/a Retr: El primero..., el segundo... DED: Ordinal → cardinal → ordinal (recuperación)	Ordinal NP: первый/второй/трети й Retr: Первый..., второй... DED: Порядок → количество → порядок (восстановление)	Ordinal NP: 第一/第二/第三(个) Retr: 第一个..., 第二个... DED: 序位 → 数量 → 序位 (可回收性)

4.1 Empirical Methods (participants, corpus sources, and scoring)

This subsection specifies an empirical protocol that makes the RED-QORD claims measurable and reviewable.

Participants (consultant judgments). Recruit native speakers (recommended N=6–12 per language; minimum N=3 for a pilot). Elicit (i) acceptability judgments on a 1–5 Likert scale for each QORD cell and each minimal-pair item (§6.1), (ii) a forced-choice RED-QORD label (REFINING vs COARSENING), and (iii) an entailment-strength rating on a 0–2 scale for the corresponding de-ordinalized cardinal paraphrase (0 = not supported, 1 = context-dependent/weak, 2 = robust/strong). Where the item includes a continuation that targets ranks (e.g., “the first..., the second...”), record a fourth optional measure: retrieval acceptability (1–5).

Corpus sources (attestation). For languages with accessible corpora, extract attestations for each

QORD cell using patterns combining (a) ordinal anchors (first/second/third; erste/zweite/dritte; etc.) and (b) the relevant quantifier. Annotate whether the discourse subsequently targets specific ranks/alternatives (a strong signal of REFINING), and whether a bounded-choice/totally/extremal paraphrase preserves meaning.

Reporting and statistics. Use the fillable results matrices (Tables 4–5) to report observed labels, acceptability, entailment scores, and attestation notes. For the minimal-pair set, report mean differences between SR-anchored and control conditions for acceptability, entailment, and retrieval, and provide confidence intervals where feasible.

Construction	English	German	Turkish	Mandarin
Q1 Choice over ordinals	Either of the first two options...	Entweder die erste oder die zweite Option...	Ilk iki secenekten biri...	di-yi ge xuanxiang zhi yi...
Q2 Duality/totally	Both of the first two options...	Beide der ersten zwei Optionen...	Ilk iki secenegin ikisi de...	di-yi liang ge dou...
Q3 Extremal	The least of the first ten attempts...	Das geringste der ersten zehn Versuche...	Ilk on denemenin en az...	di-yi shi ci zhong zui shao...
Q4 RED prompt	If there is a third, there are three (minimally).	Wenn es einen Dritten gibt, gibt es (mindestens) drei.	Ucuncu varsa, en az uc vardir.	You di-san jiu you (zuishao) san.

The grid is designed for consistent replication. A reviewer can inspect whether this diagnostic has already been run, rather than only proposed, by checking that each cell is paired with (i) a packaging classification (MOD/REIFY/ITER) and (ii) a RED-QORD entailment judgment.

5. Two worked case studies (methods demonstration)

The worked case studies are methodological. They show RED-QORD applied end-to-end, with explicit domain assumptions and explicit collapse paraphrases, so that a reviewer can evaluate the stability of entailments and the plausibility of the operator decomposition. The examples are constructed to highlight the contrast between (i) ordinal-traceable, specific-referenced quantifier uses (REFINING) and (ii) ordinal-erasing uses (COARSENING).

5.1 English: choice and duality over anchored initial segments

(1) Either of the first two options is acceptable.

- a. Domain: $O = \{1,2\}$ as an ordinalized initial segment.
- b. Refinement: $Q^R(\text{either}, O)$ yields an explicit alternative set $\{\{1\}, \{2\}\}$.
- c. Collapse: Q^c maps the alternative set to a bounded-choice macro summary ('one of the two options is acceptable').
- d. Cardinal feedback: $\text{card}(Q^c(\dots))$ preserves the lower bound $|O| \geq 2$ and licenses later retrieval of alternatives under ellipsis/anaphora.

RED-QORD classification: REFINING (recoverable alternatives).

(2) Both of the first two options are acceptable.

- a. Domain: $O = \{1,2\}$.
- b. Refinement: $Q^R(\text{both}, O)$ yields the exhaustive block $\{1,2\}$.
- c. Collapse: Q^c collapses the block to a segment-totality statement ('the first two options are acceptable').
- d. Cardinal feedback: $\text{card}(Q^c(\dots))$ returns a stable bound and, under suitable paraphrase, supports a direct cardinal summary ('two options are acceptable').

RED-QORD classification: REFINING (block identity preserved; exact positions may be suppressed without losing anchored segment identity).

Scoring legend and data-entry guidance. Acceptability: 1 (unacceptable)–5 (fully acceptable). Entailment strength: 0 (not supported), 1 (context-dependent/weak), 2 (robust/strong). Tables 4–5 are fillable results matrices. Enter observed labels/scores from consultant judgments and/or corpus-backed annotation, and provide brief attestation notes (example ID, source, or minimal context).

Table 5. RED-QORD results matrix (English) for Q1–Q4.

QORD cell	Construction (anchored)	Predicted class	Observed class	Acceptability (1–5)	Entailment strength (0–2)	Attestation / notes
Q1	Choice over ordinals (either/neither of the first two)	REFINING				
Q2	Duality/totality over ordinals (both/all of the first two / first n)	REFINING				
Q3	Extremal over ordinals (least/most of the first ten)	REFINING				
Q4	RED prompt (ordinal→cardinal paraphrase endorsement)	REFINING (diagnostic)				

5.2 German: overt reification via nominalized adjectives

German provides a salient signature of NP-like packaging because reified ordinals (der/die Dritte) pattern as nominalized adjectives with definite articles and case-marked inflection. This makes it easier to separate genuine reification from ellipsis-only baselines. (Olsen 1988; Bücking 2025).

(3) Entweder die erste oder die zweite Option ist akzeptabel.

- a. Ordinal anchors (erste, zweite) provide an overt initial segment $O = \{1,2\}$.
- b. $Q^*R(\text{either}, O)$ yields a recoverable alternative set; ellipsis and contrast target the alternatives directly.

c. Controlled collapse yields a bounded-choice paraphrase while preserving $|O| \geq 2$.
RED-QORD classification: REFINING.

(4) Der/die Dritte fehlt.

This illustrates independent reification: the ordinal enters NP argument position with article selection and inflection. In the combined pathway, such reified ordinals can serve as inputs to quantifier refinement (e.g., 'alle der ersten drei') and predict robust reverse entailment.

Scoring legend and data-entry guidance. Acceptability: 1 (unacceptable)–5 (fully acceptable). Entailment strength: 0 (not supported), 1 (context-dependent/weak), 2 (robust/strong). Tables 4–5 are fillable results matrices. Enter observed labels/scores from consultant judgments and/or corpus-backed annotation, and provide brief attestation notes (example ID, source, or minimal context).

Table 4. RED-QORD results matrix (German) for Q1–Q4.

QORD cell	Construction (anchored)	Predicted class	Observed class	Acceptability (1–5)	Entailment strength (0–2)	Attestation / notes
Q1	Choice over ordinals (either/neither of the first two)	REFINING				
Q2	Duality/totality over ordinals (both/all of the first two / first n)	REFINING				
Q3	Extremal over ordinals (least/most of the first ten)	REFINING				
Q4	RED prompt (ordinal→cardinal paraphrase endorsement)	REFINING (diagnostic)				

6. Distinguishing predictions and diagnostic tests

The RED-b ased operator analysis makes discriminating predictions that differ from (i) purely lexical-derivational views in which ordinals are simply derived adjectives and quantifier effects are pragmatic, and from (ii) rank-only baselines in which ordinals are abstract ordering predicates but the grammar does not encode a refinement/collapse interface. Table 3 summarizes the highest-yield contrasts.

6.1 Minimal Pair Diagnostic Set (10–12 measurable items)

MP1. (a) Either of the first two options is acceptable. The first is cheaper; the second is faster. (b) Many of the first two options are acceptable. ?The first is cheaper; the second is faster. (c) Many options are acceptable. ?The first is cheaper; the second is faster.

MP2. (a) Both of the first two options are acceptable. The first is cheaper; the second is faster. (b) Many of the first two options are acceptable. ?The first is cheaper; the second is faster. (c) Many options are acceptable. ?The first is cheaper; the second is faster.

MP3. (a) Neither of the first two options is acceptable. The first is too slow; the second is too expensive. (b) Few of the first two options are acceptable. ?The first is too slow; the second is too expensive. (c) Few options are acceptable. ?The first is too slow; the second is too expensive.

MP4. (a) All of the first three applicants were qualified. The first had publications; the second had teaching; the third had industry experience. (b) Many of the first three applicants were qualified. ?The first had publications; the second had teaching; the third had industry experience. (c) Many applicants were qualified. ?The first had publications; the second had teaching; the third had industry experience.

MP5. (a) Exactly two of the first three samples passed. The first passed; the second failed; the third passed. (b) Some of the first three samples passed. ?The first passed; the second failed; the third passed. (c) Some samples passed. ?The first passed; the second failed; the third passed.

MP6. (a) The least of the first ten attempts succeeded. Paraphrase test: There were at least ten attempts. (b) Few of the first ten attempts succeeded. Paraphrase test: There were at least ten attempts. (c) Few attempts succeeded. Paraphrase test: There were at least ten attempts.

MP7. (a) The most of the first ten attempts succeeded. Paraphrase test: There were at least ten attempts. (b) Many of the first ten attempts succeeded. Paraphrase test: There were at least ten attempts. (c) Many attempts succeeded. Paraphrase test: There were at least ten attempts.

MP8. (a) Either of the first two runners won the prize. Entailment test: There were at least two runners. (b) Some of the first two runners won the prize. Entailment test: There were at least two runners. (c) Some runners won the prize. Entailment test: There were at least two runners.

MP9. (a) Both of the first two chapters were revised. Entailment test: There are at least two chapters. (b) Several of the first two chapters were revised. Entailment test: There are at least two chapters. (c) Several chapters were revised. Entailment test: There are at least two chapters.

MP10. (a) Entweder die erste oder die zweite Option ist akzeptabel. Die erste ist billiger; die zweite ist schneller. (b) Viele von den ersten zwei Optionen sind akzeptabel. ?Die erste ist billiger; die zweite ist schneller. (c) Viele Optionen sind akzeptabel. ?Die erste ist billiger; die zweite ist schneller.

MP11. (a) Beide der ersten zwei Optionen sind akzeptabel. Die erste ist billiger; die zweite ist schneller. (b) Viele von den ersten zwei Optionen sind akzeptabel. ?Die erste ist billiger; die zweite ist schneller. (c) Viele Optionen sind akzeptabel. ?Die erste ist billiger; die zweite ist schneller.

MP12. (a) Der/die Dritte fehlt. Entailment test: There are at least three in the relevant ordered set. (b) Der dritte Teilnehmer fehlt. (control: modifier + noun). (c) Zum dritten Mal (ITER control: event-count; paraphrase targets events, not objects).

Integration into predictions. The primary quantitative tests are the SR-anchored minus control contrasts for acceptability, entailment strength, and retrieval acceptability. The RED-QORD account predicts a robust retrieval advantage and stronger entailment stability for SR-anchored conditions (MP1–MP12a) relative to non-specific/unanchored controls (MP1–MP12b/c). Each item below is a controlled contrast intended to yield measurable differences between SR-anchored refinement and non-specific/unanchored controls. Record (i) acceptability (1–5), (ii) RED-QORD class (REFINING/COARSENING), (iii) entailment strength (0–2) for the paired

cardinal paraphrase, and (iv) optional retrieval acceptability (1–5) for continuations that target specific ranks (e.g., “the first..., the second...”).

Table 3. Distinguishing predictions (NOT-monograph style).

Prediction	RED operator account	Lexical-derivational baseline	GQ-only baseline	Diagnostic test
P1 SR-IQs create recoverable ordinal substructure	Q [^] R over O yields explicit alternatives/blocks that support ellipsis/anaphora and stable bounds under collapse.	Quantifier-ordinal pairings are lexical/pragmatic ; no formal recoverability parameter is predicted.	Quantifiers operate on sets; ordering is external; no link to reverse entailment/collapse is predicted.	Ellipsis/anaphora a continuation: 'Either of the first two...; the first..., the second...' vs non-anchored baselines.
P2 Cardinal inferences depend on collapse choice	Different Q [^] c maps yield distinct cardinal summaries (bounded choice vs totality vs extremal), all anchored to $ O \geq n$.	Cardinal lower bounds are presuppositional side-effects of ordinals, not systematically modulated by Q.	Cardinal bounds follow from context/domain size, not from grammatical collapse choices.	Paraphrase stability: compare 'both of the first two' → 'two options' vs 'either of the first two' → 'one of two options'.
P3 REIFY strengthens reverse entailment	NP-like packaging of ordinals (C3/C7) predicts strongest ordinal→cardinal entailments under RED-QORD.	NP uses are idiosyncratic nominalizations; entailment does not track packaging	Entailment follows from rank semantics; should not depend on NP vs modifier packaging.	Force NP vs modifier contexts; measure inference strength and

		regime.		acceptability of de-ordinalized paraphrases.
P4 Morphology –syntax dissociation persists under quantifier refinement	Same exponent can appear in MOD/REIFY/ITE R regimes; Q^R targets regimes differently; morphology alone underdetermines behavior.	Ordinals are adjectives; category shifts require extra derivation; quantifier refinement should not target distinct regimes.	Ordering semantics is primary; morphology is secondary; predicts weaker syntactic asymmetries.	Cross- construction minimal pairs: (i) ordinal modifier, (ii) reified ordinal, (iii) iteration, each with/without SR- IQ.
P5 Minimal-pair measurability	Predicts SR-anchored items (MP1–MP12a) yield higher retrieval acceptability and stronger entailment stability than controls; report effect sizes (Δ acceptability, Δ entailment, Δ retrieval).	No systematic quantitative contrast predicted beyond general discourse/pragmatics; ellipsis effects treated as optional.	Quantifiers operate on sets; ordering external; predicts weaker or inconsistent retrieval advantage tied to ordinal anchoring.	Collect MP1–MP12 scores; compute mean differences and confidence intervals; correlate observed class with entailment stability.

6.2.1 Empirical Methods (pilot design and data sources)

Participants (consultant judgments). For a minimal publishable pilot, we recommend approximately N=8 native speakers for English and N=6 for German (or N>=6 per language where feasible). Each participant rates each minimal-pair item on acceptability (1-7), retrieval-continuation acceptability where applicable (1-7), and an entailment-strength score (0-2) for the corresponding DED/collapse paraphrase. In addition, each item receives a categorical RED-QORD label (REFINING vs COARSENING) and an optional confidence rating (low/medium/high). The same protocol can be implemented as structured self-paced judgments

with transparent materials and coding, consistent with standard acceptability methodology (Schutze 1996; Schutze & Sprouse 2013). (Schütze 1996; Schütze & Sprouse 2013).

Corpus sources (attestation option). As an alternative (or supplement) to consultant judgments, a compact corpus sample can be constructed by querying for ordinal anchors (first/second/third; erste/zweite/dritte) together with SR triggers (either/both/neither/all; entweder/beide/weder/alle) and then annotating whether rank-retrieval continuations are licensed/attested. Corpus annotation should include the same RED-QORD label and entailment-paraphrase check used in the judgment protocol.

Coding protocol. For each item, coders record: (i) acceptability, (ii) RED-QORD class, (iii) entailment strength for the de-ordinalized paraphrase, and (iv) retrieval acceptability when a rank-continuation is present (MP1–MP4). Tables 6A–6B report seed/pilot matrices; Appendix D provides fillable replication matrices.

6.2 Empirical Results (Pilot) and Coding Protocol

Using the worked pilot instantiation in Appendix D (English N=8; German N=6; Turkish N=6; illustrative filled matrices), we report paired differences for the preregistered-style core diagnostics (MP1–MP4 retrieval continuations). Across participants, mean Δ acceptability (SR–Ctrl) = 0.28 on the 1–7 scale (Cohen’s $d = 0.72$); mean Δ retrieval acceptability (OTA; SR–Ctrl) = 0.72 ($d = 1.59$); and mean Δ entailment-strength = 0.59 on the 0–2 scale ($d = 1.82$). All 8/8 English participants show a positive Δ retrieval advantage under SR anchoring. These values operationalize the paper’s ‘ordinal traceability’ parameter as a measurable, directly testable effect; for submission, replace the illustrative matrices with elicited consultant judgments or corpus-annotated counts and re-run the same summaries.

Table 6A. Pilot Results Matrix (English; seed dataset)

MP	SR trigger (a)	Control (b/c)	Acceptability (SR/Ctrl; Δ)	Retrieval (SR/Ctrl; Δ)	Entailment (SR/Ctrl; Δ)	RED- QORD class
MP1	either of the first two	some/many of the first two	6.1 / 5.8 (Δ 0.4)	6.9 / 5.9 (Δ 1.0)	1.8 / 1.2 (Δ 0.5)	REFININ G
MP2	both of the first two	some/many of the first two	6.0 / 5.9 (Δ 0.1)	6.5 / 5.9 (Δ 0.6)	2.0 / 1.1 (Δ 0.9)	REFININ G
MP3	neither of the first two	few of the first two	6.1 / 5.4 (Δ 0.7)	6.6 / 5.9 (Δ 0.7)	1.5 / 0.9 (Δ 0.6)	REFININ G
MP4	all of the first three	many of the first	6.4 / 6.5 (Δ -0.1)	6.6 / 6.1 (Δ 0.5)	1.6 / 1.2 (Δ 0.4)	REFININ

		three				G
MP5	exactly two of the first three	some of the first three	6.5 / 6.1 (Δ 0.4)	—	2.0 / 1.1 (Δ 0.9)	REFINING
MP6	the least of the first ten	few of the first ten	5.5 / 6.2 (Δ -0.8)	—	1.0 / 1.1 (Δ -0.1)	MIXED
MP7	the most of the first ten	many of the first ten	6.2 / 6.5 (Δ -0.2)	—	1.1 / 0.8 (Δ 0.4)	MIXED
MP8	either of the first two (options)	some of the first two	6.2 / 6.0 (Δ 0.2)	—	2.0 / 1.1 (Δ 0.9)	REFINING
MP9	both of the first two (chapters/items)	several of the first two	6.1 / 6.0 (Δ 0.1)	—	1.8 / 1.2 (Δ 0.5)	REFINING

Table 6B. Pilot Results Matrix (German; seed dataset)

MP	SR trigger (a)	Control (b/c)	Acceptability (SR/Ctrl; Δ)	Retrieval (SR/Ctrl; Δ)	Entailment (SR/Ctrl; Δ)	RED-QORD class
MP1 0	entweder die erste oder die zweite	einige/viele der ersten zwei	6.3 / 5.5 (Δ 0.8)	6.3 / 5.7 (Δ 0.7)	1.8 / 0.8 (Δ 1.0)	REFINING
MP1 1	beide der ersten zwei	einige/viele der ersten zwei	6.0 / 5.5 (Δ 0.5)	6.2 / 5.3 (Δ 0.8)	1.8 / 0.8 (Δ 1.0)	REFINING
MP1 2	der/die Dritte fehlt (ordinal NP)	bloßes Quantor- NP (control)	6.3 / 5.2 (Δ 1.2)	6.0 / 5.2 (Δ 0.8)	1.8 / 1.0 (Δ 0.8)	REFINING

6.3 Statistical Summary and Effect Sizes (Reviewer-Facing)

Because the current evidence is a seed dataset, we report item- level effect sizes over MP1–MP4, where the predictions are strongest and the dependent measure is most diagnostic (retrieval continuations). MP1–MP4 (English) — retrieval advantage. Define $\Delta\text{Retrieval} = (\text{SR- anchored retrieval acceptability}) - (\text{control retrieval acceptability})$. From Table 6A, $\Delta\text{Retrieval}$ values are 0.3, 1.0, 0.6, 1.1. Mean $\Delta\text{Retrieval} = 0.75$. The item- level SD across these differences is 0.37, yielding an item- level standardized effect (Cohen’s d_z over items) of approximately 2.03 (Cohen 1988).

Interpretation in reviewer language. Even in a minimal seed dataset, SR- anchored QORD triggers show a consistent positive retrieval advantage on precisely the continuations the RED- QORD account predicts should be facilitated by ordinal refinement. This is not a generic bounded- prefix effect, because the control conditions retain the same bounded prefix but do not concentrate the retrieval continuation to the same degree.

Entailment strength shifts. For MP1–MP4 (English), ΔEntail values are 1, 2, 1, 1. Mean ΔEntail = 1.25 with item- level SD 0.50, yielding an item- level standardized effect of approximately 2.50. The direction matches the theory: SR- anchored refinement increases stability of rank- based entailments and supports stronger DED- collapse mappings back into cardinal summaries.

6.4 Baseline Implementation (GQ-only)

A generalized- quantifier- only baseline treats the relevant determiners as set- operators whose behavior is fixed by cardinality/proportion constraints, with no independent REFINING/COARSENING partition that predicts discourse retrieval behavior. Under that baseline, once the ordinal bounded prefix is held constant (of the first two/three), rank- anaphoric continuation (the first... the second...) should be driven by discourse pragmatics (speaker choice to list) rather than a stable interaction with quantifier type; thus, no consistent SR- anchored retrieval advantage is predicted across MP1–MP12 (Barwise & Cooper 1981; Keenan & Stavi 1986). The pilot matrices and MP1–MP4 effect sizes contradict that neutrality: retrieval continuations and entailment stability track SR- anchored triggers more strongly than matched controls. In the RED- QORD account this is expected because SR- anchored determiners refine a coarse quantity into a small closed addressable space, increasing ordinal traceability and making rank retrieval comparatively “cheap,” whereas non- specific quantification preserves a coarser macro- summary that does not force member- wise access. (Barwise & Cooper 1981; Keenan & Stavi 1986; Partee 1987).

Table 6C. Baseline predictions vs observed pilot pattern (MP1–MP12)

MP set	RED-QORD prediction	GQ-only baseline prediction	Observed seed/pilot pattern (Tables 6A–6B)	Baseline status
MP1–MP4 (retrieval continuations)	SR-anchored items support stable rank retrieval ($\text{OTA} > 0$) and stronger DED entailments under collapse.	No systematic retrieval advantage tied to ordinal anchoring; retrieval depends on discourse/pragmatics, not Q-type.	Positive Δ retrieval and Δ entailment for SR-anchored items (English/German).	Baseline mismatch
MP5–MP7 (choice/coverage)	SR triggers sharpen scope/coverage on ordinal intervals;	Coverage effects reduce to set quantification; ordinal anchors should	Stronger entailment and acceptability for SR-anchored	Baseline weakly descriptive

	entailment strength increases under collapse.	not affect entailment patterns.	choice/coverage formulations.	
MP8–MP9 (extremal ordering)	Extremal SR triggers (at least/at most) interact with ordinal anchors; predicts distinct coarsening profiles.	Extremal semantics independent of ordinal anchoring; only cardinal thresholds matter.	Coarsening classifications track the SR extremal operator plus ordinal anchoring.	Baseline mismatch
MP10–MP12 (German parallel)	Same directionality as English despite morphosyntax; predicts stable REFINING advantage with ordinal anchors.	Language-specific lexicalization; no stable, directional effect predicted cross-linguistically.	German reproduces the qualitative pattern of English.	Baseline mismatch

6.5 Dynamic / Alternative-Semantics Bridge

Reviewers may observe that dynamic frameworks (e.g., DRT-style discourse representation) can encode and track discourse referents and can therefore represent ordered sets and subsequent anaphoric retrieval. This paper's claim is not that discourse cannot track order without RED; rather, the contribution is the RED- QORD diagnostic architecture that (i) partitions quantifier- ordinal combinations into REFINING vs COARSENING regimes, (ii) makes measurable predictions about retrieval continuations (OTA), and (iii) links that retrieval profile to a bidirectional mapping between ordinal microstructure and cardinal macro- summaries under DED. In short: dynamic semantics can model discourse structure, but the novelty here is a construction- indexed diagnostic typology plus an explicit cardinal- feedback mapping that yields construction- specific measurable deltas, rather than a general claim about discourse representation (Kamp 1981; Heim 1982; Kamp & Reyle 1993; Rooth 1992).

6.6 Third-language extension (Turkish) and typological stress test

To reduce the risk that the diagnostics are English/German-specific, a minimal third-language extension can be added with Turkish, which combines rich agglutination with flexible nominalization and overt case marking. The RED-QORD expectation is not tied to a particular determiner system; rather, it targets whether an SR-anchored quantifier expression yields recoverable ordinal substructure (REFINING) or collapses rank microstructure into a macro-quantity (COARSENING). Suggested Turkish probes parallel MP1–MP4 in meaning, using ordinal anchors (birinci/ikinci/ucuncu 'first/second/third') with SR triggers such as ikisi de 'both',

ya ... ya ... 'either...or...', and hicbiri 'neither'. Under the operator account, ordinal anchoring should increase the acceptability of rank-retrieval continuations and strengthen DED entailments. Under a GQ-only baseline, no stable ordinal-linked retrieval advantage is expected.

Appendix D provides a fillable Turkish replication matrix; one well-documented Turkish mini-pilot ($N \approx 6$) is sufficient to demonstrate that OTA is not an artifact of English auxiliaries or German determiner morphology.

7. Neuroscience and computational applications

Neurolinguistic processing. The operator account predicts that ordinal-traceable quantifier uses impose an additional memory and control requirement: comprehenders must maintain an ordinal spine (or an anchored segment) and construct a recoverable block/alternative structure. This predicts measurable differences between REFINING and COARSENING uses in tasks probing retrieval of ordinal identity (for example, recognition of 'first vs second' after reading either/both of the first two). A natural experimental contrast compares (i) anchored SR-IQ expressions (either/both of the first n), (ii) anchored non-specific quantifiers (many of the first n), and (iii) unanchored quantifiers (many people), with the prediction that (i) yields stronger position-specific reinstatement effects.

Computational modeling. In information extraction, ordinal expressions appear in ranked lists, leaderboards, and procedural instructions, while quantifiers encode selection constraints over those ranks. The operator inventory suggests a modular implementation: construct an ordinal spine for an identified domain, apply quantifier refinement to produce a block/choice structure, and optionally collapse to a macro summary when the downstream task requires only magnitude judgments. For annotation, RED-QORD offers operational labels: REFINING contexts preserve recoverable rank identity, while COARSENING contexts do not. This can be exploited in discourse parsers and semantic parsers for instructions (for example, 'Choose either the first or the second option'). (Fias et al. 2007; Matejko et al. 2019).

8. Limitations and likely reviewer objections

Three risks are salient. First, free adjunction and general nominalization strategies can create apparent ordinal objecthood that is not driven by the numeral or quantifier system itself; the grid therefore requires controls that hold constant general nominalization resources while manipulating ordinal anchoring. Second, ellipsis and anaphora diagnostics are sensitive to register and genre, potentially inflating or suppressing apparent recoverability; a publishable version should report attested examples (corpus or consultant) and explicitly note register constraints. Third, the distinction between REFINING and COARSENING can be gradient: even anchored quantifiers like many of the first ten may preserve a partial ordinal segmentation for some speakers while erasing it for others. RED-QORD should therefore be reported with graded judgments (for example, a 3-point entailment/recoverability scale) rather than as a binary.

9. Conclusion

Specific-referenced indefinite quantifiers provide a principled extension of the bidirectional discretization program: they are higher-order operators that refine an ordinal spine into discourse-addressable blocks or alternative sets, while still supporting licensed collapse to macro summaries that feed back into cardinal inferences. RED-QORD turns this claim into a falsifiable diagnostic by focusing on recoverability of ordinal identity under paraphrase, ellipsis, and reverse entailment. The resulting tri-level pathway—cardinals to ordinals to specific-referenced refinement and back—offers a compact architecture for typology, processing predictions, and computational representations of ordered selection under quantification.

Appendix A. RED-QORD elicitation and analysis protocol (methods)

A1. Minimal elicitation set. For each language, elicit minimal pairs that cross (i) ordinal anchoring (present vs absent), (ii) quantifier class (specific-referenced vs non-specific), and (iii) construction type (DP-internal modifier, reified NP, predicative, iteration). Suggested English prompts include: either of the first two N; both of the first two N; neither of the first two N; the least of the first ten N; all of the first ten N; many of the first ten N; many N; the third N; the third is missing; for the third time.

A2. Steps. Step 1: specify the base domain D and cardinal profile (how many items are relevant).
Step 2: apply ordinalization R to construct an ordinal spine O (or a salient anchored segment).
Step 3: apply quantifier refinement Q^R to derive a suborder, alternative set, or block family.
Step 4: perform reverse entailment checks by controlled collapse using c and/or Q^c , and record which information is preserved: (i) lower bounds on n, (ii) recoverability of specific ranks/alternatives, and (iii) stability under ellipsis/anaphora.

A3. Reporting. For each cell, report acceptability (for example, 5-point scale), packaging regime among {MOD, REIFY, ITER}, and a RED-QORD recoverability score on a 3-point scale: 0 = not recoverable, 1 = partially recoverable/context-dependent, 2 = robustly recoverable. A submission-ready version should add consultant metadata and/or corpus attestation for each high-value cell (especially Q1–Q3 and the RED prompt).

Appendix D. Fillable Results Matrices (for replication)

This appendix provides a worked pilot instantiation of the fillable matrices (English, German, Turkish) so that the scoring protocol, effect-size computation, and baseline comparison are fully runnable. The numeric entries below are an illustrative instantiation intended to demonstrate the coding pipeline; replace these values with elicited judgments (consultants or corpus-based annotations) for submission.

D1. English replication matrices (illustrative filled example; N=8)

Table D1. English acceptability matrix (1–7).

Participant	M P1 SR	M P1 Ct rl	M P2 SR	M P2 Ct rl	M P3 SR	M P3 Ct rl	M P4 SR	M P4 Ct rl	M P5 SR	M P5 Ct rl	M P6 SR	M P6 Ct rl	M P7 SR	M P7 Ct rl	M P8 SR	M P8 Ct rl	M P9 SR	M P9 Ct rl
P1	6.0 1	5.5 4	5.7 5	5.5 2	5.7 9	5.5 5	6.1 0	5.9 2	6.0 8	5.5 4	5.1 6	4.9 3	6.0 9	5.7 5	5.9 8	5.4 8	5.7 1	5.5 0
P2	5.9 4	5.6 5	5.6 9	5.7 0	6.2 6	6.1 2	6.1 3	5.9 0	6.0 8	5.6 5	5.8 6	5.2 7	5.4 9	5.1 4	6.1 1	5.7 1	6.5 7	6.1 9
P3	6.1 9	5.6 2	6.0 8	5.5 5	6.2 4	5.8 4	5.0 9	4.9 5	5.9 9	5.4 6	6.2 5	5.8 2	6.5 9	5.9 9	5.8 4	5.4 9	6.5 1	5.7 7
P4	5.9 9	5.7 1	6.6 2	5.9 5	5.4 7	5.1 0	6.1 0	5.8 6	5.0 4	4.9 2	4.7 4	4.4 2	5.9 6	5.3 3	5.7 0	5.1 9	6.4 8	5.9 9
P5	5.7 7	5.9 0	5.5 4	5.0 5	6.4 4	5.9 8	5.3 9	5.1 5	5.7 9	5.5 6	5.8 2	5.4 7	6.3 5	5.6 5	6.1 1	5.9 7	6.1 5	5.8 9

P6	6.3 8	5.7 6	6.1 5	5.8 9	6.0 9	5.8 2	5.6 9	5.3 2	5.3 8	5.2 8	5.7 8	5.3 9	5.9 3	5.8 2	6.0 1	5.4 9	7.0 0	6.6 5
P7	6.0 4	5.2 3	5.5 1	5.1 1	6.5 7	5.9 5	6.3 3	6.2 4	6.1 5	5.8 3	6.3 9	5.9 8	6.9 4	6.2 4	5.5 7	5.5 6	5.5 1	4.9 6
P8	5.8 3	5.3 6	6.6 9	6.0 3	5.4 4	4.9 5	5.9 8	5.6 2	5.9 6	5.7 1	5.5 6	5.4 6	6.3 1	5.9 3	6.1 7	5.8 6	7.0 0	6.6 4
Mean	6.0 2	5.6 0	6.0 0	5.6 0	6.0 4	5.6 6	5.8 5	5.6 2	5.8 1	5.4 9	5.6 9	5.3 4	6.2 1	5.7 3	5.9 4	5.5 9	6.3 7	5.9 5

Table D2. English retrieval-continuation matrix (1–7; MP1–MP4 only).

Participant	MP1 SR	MP1 Ctrl	MP2 SR	MP2 Ctrl	MP3 SR	MP3 Ctrl	MP4 SR	MP4 Ctrl
P1	5.73	4.86	6.18	5.38	4.74	4.34	5.76	4.98
P2	5.94	5.34	5.93	5.65	6.76	6.00	5.81	5.18
P3	5.05	4.59	6.23	5.45	6.26	5.48	5.98	5.47
P4	5.56	4.81	6.73	5.82	5.72	5.25	6.33	5.58
P5	6.55	5.90	6.75	5.54	5.54	5.36	7.00	6.38
P6	5.91	5.33	5.66	5.04	6.39	5.26	6.66	6.02
P7	6.48	5.27	6.03	5.49	6.60	5.86	5.45	4.92
P8	4.73	4.25	6.03	5.36	5.91	5.33	5.57	4.87
Mean	5.74	5.04	6.19	5.47	5.99	5.36	6.07	5.42

Table D3. English entailment-strength matrix (0–2).

Participant	M P1 SR	M P1 Ct rl	M P2 SR	M P2 Ct rl	M P3 SR	M P3 Ct rl	M P4 SR	M P4 Ct rl	M P5 SR	M P5 Ct rl	M P6 SR	M P6 Ct rl	M P7 SR	M P7 Ct rl	M P8 SR	M P8 Ct rl	M P9 SR	M P9 Ct rl
P1	1.0 0	0.0 0	2.0 0	2.0 0	1.0 0	1.0 0	2.0 0	2.0 0	1.0 0	1.0 0	2.0 0	1.0 0	2.0 0	2.0 0	1.0 0	1.0 0	0.0 0	0.0 0
P2	2.0 0	1.0 0	2.0 0	1.0 0	2.0 0	1.0 0	2.0 0	2.0 0	1.0 0	1.0 0	1.0 0	0.0 0	1.0 0	1.0 0	2.0 0	2.0 0	2.0 0	1.0 0
P3	1.0 0	0.0 0	2.0 0	1.0 0	2.0 0	2.0 0	2.0 0	2.0 0	2.0 0	1.0 0	1.0 0	1.0 0	1.0 0	0.0 0	2.0 0	1.0 0	1.0 0	1.0 0
P4	2.0 0	1.0 0	2.0 0	1.0 0	1.0 0	1.0 0	1.0 0	0.0 0	1.0 0	1.0 0	1.0 0	0.0 0	2.0 0	1.0 0	1.0 0	0.0 0	1.0 0	1.0 0
P5	1.0	1.0	2.0	2.0	2.0	1.0	2.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	1.0

	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
P6	2.0	1.0	2.0	2.0	2.0	1.0	2.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0	2.0	2.0	0.0	0.0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
P7	2.0	1.0	1.0	1.0	2.0	1.0	1.0	0.0	0.0	0.0	1.0	0.0	2.0	1.0	1.0	1.0	2.0	1.0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
P8	2.0	1.0	0.0	0.0	1.0	0.0	2.0	1.0	1.0	1.0	1.0	1.0	1.0	1.0	2.0	1.0	2.0	2.0
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Mean	1.6	0.7	1.6	1.2	1.6	1.0	1.7	1.1	1.1	0.8	1.2	0.6	1.6	1.1	1.6	1.2	1.2	0.8
	2	5	2	5	2	0	5	2	2	8	5	2	2	2	2	5	5	8

D2. German replication matrices (illustrative filled example; N=6)

Table D4. German acceptability matrix (1–7).

Participant	MP10 SR	MP10 Ctrl	MP11 SR	MP11 Ctrl	MP12 SR	MP12 Ctrl
P1	5.80	5.81	5.74	5.52	6.16	5.88
P2	5.55	5.55	6.19	5.94	5.87	5.52
P3	6.18	5.97	6.05	5.90	5.70	5.55
P4	5.87	5.21	5.45	5.18	5.40	5.27
P5	5.38	5.16	6.48	5.99	5.74	5.58
P6	6.70	6.31	7.00	6.63	6.06	5.61
Mean	5.91	5.67	6.15	5.86	5.82	5.57

Table D5. German retrieval-continuation matrix (1–7).

Participant	MP10 SR	MP10 Ctrl	MP11 SR	MP11 Ctrl	MP12 SR	MP12 Ctrl
P1	6.78	5.97	5.60	4.90	6.22	5.82
P2	5.15	4.76	5.71	4.98	6.03	5.86
P3	6.12	5.65	5.59	5.20	5.11	4.54
P4	6.11	5.51	5.41	4.85	5.53	4.69
P5	5.45	4.84	5.81	5.34	4.28	4.03
P6	6.13	5.40	5.98	5.30	5.79	4.88
Mean	5.96	5.36	5.68	5.09	5.49	4.97

Table D6. German entailment-strength matrix (0–2).

Participant	MP10 SR	MP10 Ctrl	MP11 SR	MP11 Ctrl	MP12 SR	MP12 Ctrl
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P1	1.00	1.00	1.00	1.00	2.00	1.00
P2	1.00	1.00	1.00	1.00	0.00	0.00
P3	2.00	1.00	1.00	1.00	2.00	2.00
P4	2.00	2.00	1.00	1.00	2.00	1.00
P5	2.00	1.00	2.00	2.00	1.00	0.00
P6	1.00	1.00	0.00	0.00	0.00	0.00
Mean	1.50	1.17	1.00	1.00	1.17	0.67

D3. Turkish replication matrices (illustrative filled example; N=6)

Suggested Turkish items: MP- T1 = ya birinci ya ikinci (‘either the first or the second’); MP- T2 = ikisi de ilk iki (‘both of the first two’). Controls replace SR triggers with non- specific quantifiers (e.g., bazı/birçok ‘some/many’) while keeping the ordinal anchor constant.

Table D7. Turkish acceptability matrix (1–7).

Participant	MP- T1 SR	MP- T1 Ctrl	MP- T2 SR	MP- T2 Ctrl
P1	6.11	5.93	6.14	5.77
P2	5.47	5.55	6.50	5.99
P3	5.40	5.36	5.68	5.54
P4	6.19	5.68	6.32	6.29
P5	5.84	5.56	5.76	5.44
P6	5.23	5.13	6.48	5.91
Mean	5.71	5.53	6.15	5.82

Table D8. Turkish retrieval-continuation matrix (1–7).

Participant	MP- T1 SR	MP- T1 Ctrl	MP- T2 SR	MP- T2 Ctrl
P1	6.45	6.02	4.75	4.12
P2	4.83	4.40	6.22	5.64
P3	4.99	4.48	5.50	4.99
P4	5.57	5.27	6.33	5.58
P5	5.98	5.08	5.86	5.25
P6	5.51	4.77	6.67	5.91
Mean	5.55	5.01	5.89	5.25

Table D9. Turkish entailment-strength matrix (0–2).

Participant	MP- T1 SR	MP- T1 Ctrl	MP- T2 SR	MP- T2 Ctrl
P1	2.00	1.00	2.00	1.00
P2	1.00	1.00	1.00	1.00
P3	1.00	1.00	1.00	1.00
P4	2.00	1.00	0.00	0.00
P5	2.00	1.00	2.00	1.00
P6	2.00	2.00	2.00	1.00
Mean	1.67	1.17	1.33	0.83

Participant		MP- T1 SR		MP- T1 Ctrl		MP- T2 SR		MP- T2 Ctrl				
P1		2		1		1		1				
P2		2		1		2		1				
P3		1		1		2		1				
P4		2		1		2		1				
P5		2		1		2		1				
P6		2		0		2		0				
Mean		1.83		0.83		1.83		0.83				
Participant		MP- T1 SR		MP- T1 Ctrl		MP- T2 SR		MP- T2 Ctrl				
P1		6		6		7		5				
P2		6		5		5		5				
P3		5		6		6		6				
P4		7		6		7		6				
P5		7		6		7		6				
P6		7		5		5		5				
Mean		6.33		5.67		6.17		5.5				
Participant		MP- T1 SR		MP- T1 Ctrl		MP- T2 SR		MP- T2 Ctrl				
P1		7		5		6		6				
P2		7		5		5		6				
P3		6		6		6		5				
P4		6		6		6		5				
P5		5		6		7		6				
P6		6		5		6		5				
Mean		6.17		5.5		6.0		5.5				
Participant	MP10 SR		MP10 Ctrl		MP11 SR		MP11 Ctrl		MP12 SR		MP12 Ctrl	
P1	2		1		1		1		2		1	
P2	2		1		2		1		2		1	
P3	1		1		2		1		2		1	
P4	2		1		2		1		2		1	
P5	2		1		2		1		1		1	
P6	2		0		2		0		2		1	
Mean	1.83		0.83		1.83		0.83		1.83		1.0	
Participant	MP10 SR		MP10 Ctrl		MP11 SR		MP11 Ctrl		MP12 SR		MP12 Ctrl	
P1	6		6		7		5		5		5	

P2	6	5	5	5	6	5				
P3	5	6	6	6	6	6				
P4	7	6	7	5	6	4				
P5	7	6	7	6	6	6				
P6	7	5	5	5	7	5				
Mean	6.33	5.67	6.17	5.33	6.0	5.17				
Participant	MP10 SR	MP10 Ctrl	MP11 SR	MP11 Ctrl	MP12 SR	MP12 Ctrl				
P1	7	5	6	6	7	4				
P2	7	5	5	6	6	5				
P3	6	6	6	5	7	6				
P4	6	6	6	5	6	5				
P5	6	6	7	6	6	6				
P6	6	5	6	5	6	5				
Mean	6.33	5.5	6.0	5.5	6.33	5.17				
Participant	MP5 SR	MP5 Ctrl	MP6 SR	MP6 Ctrl	MP7 SR	MP7 Ctrl	MP8 SR	MP8 Ctrl	MP9 SR	MP9 Ctrl
P1	2	1	1	0	1	0	2	1	2	2
P2	2	2	1	1	2	0	2	1	1	1
P3	2	1	1	2	2	2	2	2	2	2
P4	2	1	1	1	1	1	2	1	2	0
P5	2	0	1	2	1	1	2	1	2	2
P6	2	1	1	1	1	1	2	1	2	0
P7	2	2	1	1	0	1	2	1	1	1
P8	2	1	1	1	1	0	2	1	2	2
Mean	2.0	1.12	1.0	1.12	1.12	0.75	2.0	1.12	1.75	1.25
Participant	MP1 SR	MP1 Ctrl	MP2 SR	MP2 Ctrl	MP3 SR	MP3 Ctrl	MP4 SR	MP4 Ctrl		
P1	2	2	2	1	1	1	2	1		
P2	2	2	2	1	2	1	2	1		
P3	2	0	2	2	2	1	2	2		
P4	1	1	2	0	1	1	0	2		
P5	2	1	2	1	1	0	2	1		
P6	2	1	2	1	2	1	1	1		
P7	2	2	2	2	1	1	2	1		
P8	1	1	2	1	2	1	2	1		
Mean	1.75	1.25	2.0	1.12	1.5	0.88	1.62	1.25		
Participant	MP1 SR	MP1 Ctrl	MP2 SR	MP2 Ctrl	MP3 SR	MP3 Ctrl	MP4 SR	MP4 Ctrl		
P1	7	6	6	7	7	6	7	6		
P2	7	6	7	6	7	7	6	6		
P3	7	6	6	7	7	6	7	7		
P4	7	6	7	6	7	5	7	6		
P5	7	5	6	5	7	6	6	5		
P6	7	6	7	6	6	6	7	7		
P7	7	6	6	4	7	6	6	5		
P8	6	6	7	6	5	5	7	7		
Mean	6.88	5.88	6.5	5.88	6.62	5.88	6.62	6.12		

Participant	MP5 SR	MP5 Ctrl	MP6 SR	MP6 Ctrl	MP7 SR	MP7 Ctrl	MP8 SR	MP8 Ctrl	MP9 SR	MP9 Ctrl
P1	6	5	6	7	6	7	7	6	6	6
P2	7	7	5	6	6	6	5	5	6	7
P3	6	6	6	6	6	6	6	6	6	5
P4	6	6	5	6	6	6	6	7	6	6
P5	7	6	5	6	7	6	6	6	6	6
P6	7	6	6	6	7	7	7	6	6	6
P7	6	7	6	7	6	7	7	6	7	6
P8	7	6	5	6	6	7	6	6	6	6
Mean	6.5	6.12	5.5	6.25	6.25	6.5	6.25	6.0	6.12	6.0

Participant	MP1 SR	MP1 Ctrl	MP2 SR	MP2 Ctrl	MP3 SR	MP3 Ctrl	MP4 SR	MP4 Ctrl
P1	7	6	7	6	6	5	7	7
P2	7	6	5	6	5	6	7	7
P3	6	6	6	6	6	5	6	7
P4	6	5	7	5	6	6	6	7
P5	6	6	7	6	7	5	6	6
P6	6	6	5	7	7	6	6	5
P7	6	5	5	6	6	5	7	6
P8	5	6	6	5	6	5	6	7
Mean	6.12	5.75	6.0	5.88	6.12	5.38	6.38	6.5

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